Netflix Movies and TV Shows Analysis Report

Analyst Profile

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Education: Final-Year B.Tech Student (Electronics & Computer Engineering)
Career Aspiration: Data Analyst
Technical Skills: Python | SQL | Power BI | Excel

This analysis was independently conducted as part of my Data Analytics portfolio to demonstrate practical skills in data cleaning, exploratory data analysis (EDA), data visualization, and business insight generation. The project follows a structured analytical workflow commonly used in industry, beginning with data preprocessing and progressing through trend analysis, visualization, and interpretation of findings.

The objective was not only to analyze Netflix's content catalog but also to derive meaningful business insights from real-world data. Through this project, I applied analytical thinking, statistical reasoning, and data storytelling techniques to transform raw data into actionable observations.

Technical Skills Applied

- **Data Cleaning & Preprocessing**
- **Exploratory Data Analysis (EDA)**
- **Data Visualization**
- **Business Intelligence & Reporting**
- **Trend Analysis**
- **Data Storytelling**
- **Python (Pandas, NumPy)**
- **Matplotlib & Seaborn**
- **Jupyter Notebook**

Executive Summary

This project analyzes Netflix's catalog of Movies and TV Shows to identify content distribution patterns, growth trends, leading content-producing countries, popular genres, and frequently featured directors and actors. Using Python-based Exploratory Data Analysis (EDA), meaningful insights were extracted from the dataset to better understand Netflix's content strategy, audience preferences, and global market reach.

The analysis revealed that Movies dominate Netflix's content library, content additions increased significantly after 2015, and countries such as the United States and India contribute substantially to the platform's catalog. Genre analysis further highlighted the popularity of Drama, Comedy, Documentary, and International content categories.

Business Objective

The primary objective of this project is to analyze Netflix's content library and identify trends related to content distribution, growth, genre popularity, country-wise contributions, and creator participation. The analysis aims to transform raw streaming platform data into meaningful business insights that can support content strategy and decision-making.

1. Introduction

Netflix is one of the world's largest streaming platforms, offering thousands of movies and television shows across various genres and countries. Understanding the composition of Netflix's content library can provide valuable insights into audience preferences, regional content production, and platform growth.

The objective of this analysis is to explore the Netflix dataset and identify key trends related to content type, content growth over time, country-wise contributions, genres, directors, and cast members.

2. Dataset Description

The dataset contains metadata about Netflix titles, including:

- **Show ID**
- **Title**
- **Type (Movie / TV Show)**
- **Director**
- **Cast**
- **Country**
- **Date Added**
- **Release Year**
- **Rating**
- **Duration**
- **Genre (Listed In)**

The dataset consists of Netflix content available up to 2019 and provides valuable information for understanding content trends and platform expansion.

3. Data Preprocessing

Before analysis, data cleaning was performed to ensure consistency and accuracy.

Missing Value Treatment

The following preprocessing steps were applied:

Column	Action Taken
Director	Replaced missing values with "No Director"

Column	Action Taken
Cast	Replaced missing values with "No Cast"
Country	Replaced missing values with "Country Unavailable"
Date Added	Rows with missing values removed
Rating	Rows with missing values removed

These preprocessing steps ensured that the dataset was complete and suitable for reliable analysis.

4. Exploratory Data Analysis

4.1 Content Distribution by Type

A comparison was performed between Movies and TV Shows available on Netflix.

Findings

- **Movies constitute the majority of Netflix's content library.**
- **TV Shows account for a smaller but significant portion of the platform.**
- **The analysis indicates that Movies represent a substantially larger share of Netflix's catalog compared to TV Shows.**

Business Insight

Netflix has built a content portfolio heavily centered around Movies, enabling broader audience reach and greater content diversity.

4.2 Growth of Netflix Content Over Time

The number of titles added each year was analyzed to understand platform growth.

Findings

- **Content additions increased significantly after 2015.**
- **The growth rate accelerated between 2016 and 2019.**
- **Both Movies and TV Shows displayed strong upward trends.**
- **Movies experienced faster growth than TV Shows.**

Business Insight

The rapid increase in content additions reflects Netflix's aggressive global expansion strategy and substantial investment in content acquisition and original productions.

4.3 Country-wise Content Production

Countries were analyzed based on the number of titles contributed to Netflix.

Findings

- **The United States emerged as the leading content-producing country.**
- **India ranked among the top contributors.**
- **The United Kingdom, Canada, and Japan also contributed significant volumes of content.**
- **Most Netflix content originates from a relatively small number of major entertainment markets.**

Business Insight

Netflix leverages content from established entertainment industries while simultaneously expanding localized content to strengthen its presence in international markets.

4.4 Top Directors on Netflix

The number of titles associated with each director was evaluated.

Findings

- **A small group of directors contributed a disproportionately large number of titles.**
- **Several directors appeared repeatedly throughout the Netflix catalog.**

Business Insight

The concentration of titles among a limited number of directors suggests strong content partnerships and recurring contributions from established creators.

4.5 Genre Analysis

Genres were examined to determine the most common content categories.

Findings

The most represented genres include:

- **International Movies**
- **Dramas**
- **Comedies**
- **Documentaries**
- **Action & Adventure**

Business Insight

Netflix maintains a diverse content portfolio while placing significant emphasis on globally popular genres such as Drama and Comedy, helping maximize audience engagement across different regions.

4.6 Most Featured Actors in TV Shows

Actor appearances within TV Shows were analyzed.

Findings

- **Certain actors appeared repeatedly across multiple television series.**
- **Recurring actors achieved greater visibility due to their presence in multiple shows and seasons.**

Business Insight

Frequent actor appearances can strengthen audience familiarity and contribute to long-term viewer retention.

4.7 Most Featured Actors in Movies

Actor appearances within Movies were analyzed separately.

Findings

- **Several actors appeared across numerous movie titles.**
- **Movie casts exhibited greater diversity compared to television content.**

Business Insight

A diverse range of actors allows Netflix to appeal to wider audience segments and support varied content offerings.

5. Key Insights

1. **Movies dominate Netflix's content catalog.**
2. **Netflix experienced rapid content growth after 2015.**
3. **The United States is the largest content contributor.**

4. **India is one of the most significant non-US content-producing countries.**
5. **Drama, Comedy, and International Movies are among the most popular genres.**
6. **Netflix collaborates extensively with a select group of directors.**
7. **Actor representation varies significantly between Movies and TV Shows.**
8. **Netflix's growth strategy relies on both international expansion and content diversification.**

6. Conclusion

The analysis reveals Netflix's strong focus on movie content, rapid platform expansion, and global content diversification. The United States remains the dominant contributor, while countries such as India play an increasingly important role in content production. Genre analysis highlights audience preference for universally appealing categories such as Drama, Comedy, and International content.

Overall, Netflix's content strategy demonstrates a balance between international expansion, genre diversity, and large-scale content acquisition, enabling the platform to maintain its position as a global leader in the streaming industry.

This project successfully demonstrates the application of data analytics techniques to a real-world dataset and highlights the value of transforming raw data into actionable business insights.

Tools & Technologies Used

- **Python**
- **Pandas**
- **NumPy**

- **Matplotlib**
- **Seaborn**
- **Jupyter Notebook**

Skills Developed Through This Project

- **Data Cleaning & Preprocessing**
- **Exploratory Data Analysis (EDA)**
- **Data Visualization**
- **Statistical Interpretation**
- **Business Insight Generation**
- **Data Storytelling**
- **Trend Analysis**
- **Python-Based Data Analytics**
- **Reporting & Presentation of Findings**

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Project Type: Exploratory Data Analysis (EDA)
Domain: Entertainment & Streaming Analytics